# The cogmod R Package: Easy-to-Use Bayesian Cognitive Models for Reaction Times, Choices and Subjective Ratings, with Applications to Computational Neuropsychology

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Cognitive tasks yield far more information than the condition means that dominate published analyses. Descriptive distributional families such as the ex-Gaussian, and process models such as the Drift Diffusion Model, both recover structure that a mean difference discards. Adoption remains limited, and the obstacle is practical: each model family lives in its own package, with its own syntax, its own parameterization and its own output objects. Families are therefore rarely compared, and cognitive modelling sits apart from the regression workflow psychologists already use. We introduce `cogmod`, an R package implementing a growing collection of such models as `brms` custom families. Models are specified with ordinary `brms` formula syntax; any parameter (drift rate, boundary separation, non-decision time, dispersion, precision) can be predicted by covariates and given random effects; and the fitted object is a standard `brmsfit`, supporting the usual priors, diagnostics, posterior predictive checks, leave-one-out cross-validation and `easystats` post-processing. Three worked examples, one per family of models, use simulated, laboratory and clinical data. Five evidence accumulation architectures are fitted to the same choices and reaction times, compared in a single call, and the misfit of the simplest one traced to a specific missing parameter. Sixteen reaction time families are then compared with the choice discarded, before the same likelihood is used to ask whether its parameters are precise enough to serve as individual-level scores, and to reanalyse a task-switching dataset in which the conventional executive-control index does not separate ADHD participants from controls at all, while the model localises the group difference in two components that cancel at the level of the mean. A third example reaches bounded rating scales through the same interface. `cogmod` is openly available at https://github.com/DominiqueMakowski/cogmod.

*Keywords:* cognitive modelling, Bayesian statistics, reaction times, evidence accumulation, subjective ratings, computational neuropsychology, R

## Introduction

### Traditional models can be misleading

Consider a dataset of reaction times (RTs) from 20 participants tested under two experimental conditions (25 trials each), available in `cogmod`. A traditional approach would involve fitting a linear mixed model testing the effect of condition, with participants entered as random intercepts:

```
lmer(RT ~ Condition + (1 | Participant),
     data = cogmod::badlm)
```

It finds no significant difference ($b = -0.001$, $t(996) = -0.08$, $p = .94$; Figure 1 B), which one could misleadingly report as an absence of any difference in the RTs of the two conditions. Yet the two distributions could hardly be more different (Figure 1 A): condition A produces a lot of very fast and very slow responses, whereas condition B yields much more consistent RTs. But because the mean of their distribution is similar, the analysis does not capture any other differences. The problem is not the design, the data preprocessing, the random-effect structure or the post-processing, but the model itself: linear regressions estimate means and nothing else. Simply changing the model to use another distribution, such as ex-Gaussian, drastically changes the conclusions that can be drawn from the same data, revealing significant differences in the bulk of the distribution, its spread and its tail (Figure 1 C). Despite the massive benefits we can get from using more appropriate models, they are rarely used in practice, which is what the `cogmod` package aims to address.

### From means to mechanisms

Cognitive tasks yield complex data, resulting from a sequence of processes that unfold over time. Some responses come almost immediately, some follow a lapse of attention, some reflect an unusually cautious pass over the stimulus, and some are emitted before the evidence has really been weighed. The shape of the distribution, its mode, its spread and its tail, is where the underlying processes leave their trace. Most published analyses reduce all of it to a handful of condition means and test for their difference. That reduction is itself a modelling decision, and a costly one: most of what makes cognitive data distinctive is discarded in the process.

The traditional summary-statistics approach involving *t*-tests, ANOVAs, linear regressions, treats observations as Normally distributed around a predicted mean (Figure 2). Two alternatives can be distinguished from it. The **distributional** approach chooses a likelihood that matches the shape of the outcome, such as an ex-Gaussian or shifted Log-Normal for reaction times (Balota & Yap, 2011; RTs, Matzke & Wagenmakers, 2009), or an ordered Beta for bounded ratings (Kubinec, 2023), so that skew, boundedness and dispersion are described rather than ignored. The **computational** approach goes further and derives the likelihood from an account of the process that generated the data. Evidence accumulation models (EAMs, or sequential sampling models) treat response times as the first-passage times of a noisy accumulation towards a decision boundary, and their parameters map onto theoretically distinct components of the decision: the quality of the evidence, how much of it is required before committing, and the time spent encoding the stimulus and executing the response (Brown & Heathcote, 2008; Ratcliff & McKoon, 2008).

The case for moving past the first approach is well supported. Distributional analyses recover effects that mean comparisons miss and disambiguate effects that mean comparisons conflate (Balota & Yap, 2011; De Boeck & Jeon, 2019). Process models turn an experimental effect into a statement about *which component of processing changed*, which is what most cognitive-theoretical claims are about (Evans & Wagenmakers, 2020; Ratcliff et al., 2016). Model-based parameters are also more reliable than the difference scores tra-

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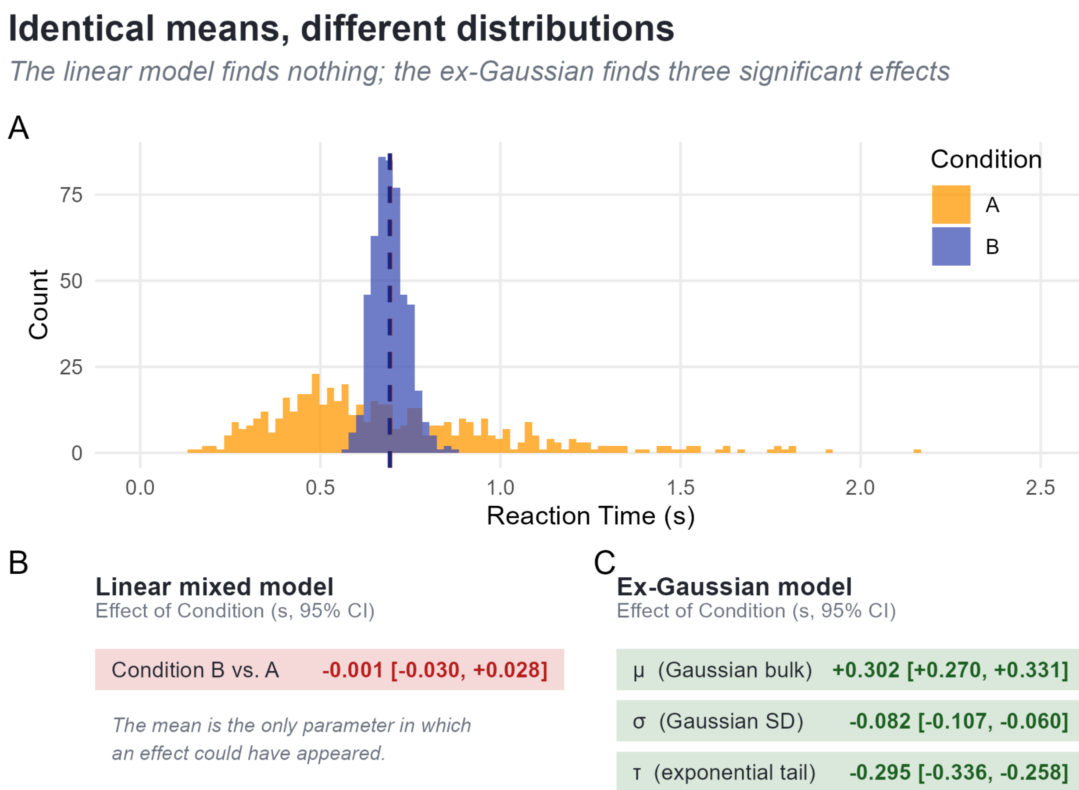
**Figure 1**

*Simulated reaction times from two experimental conditions, available in the* `cogmod` *package.* ***(A)*** *Observed distributions; dashed lines mark the condition means.* ***(B, C)*** *The effect of condition as each model reports it, with 95% intervals; green marks an interval excluding zero, red one that does not. The linear mixed model has only a mean to report, and finds no effect in it. The ex-Gaussian splits the same data into a Gaussian bulk ($\mu$), its spread ($\sigma$) and an exponential tail ($\tau$), and finds an effect on all three; because its mean is $\mu + \tau$, the opposing effects on bulk and tail cancel into exactly the null of panel B.*

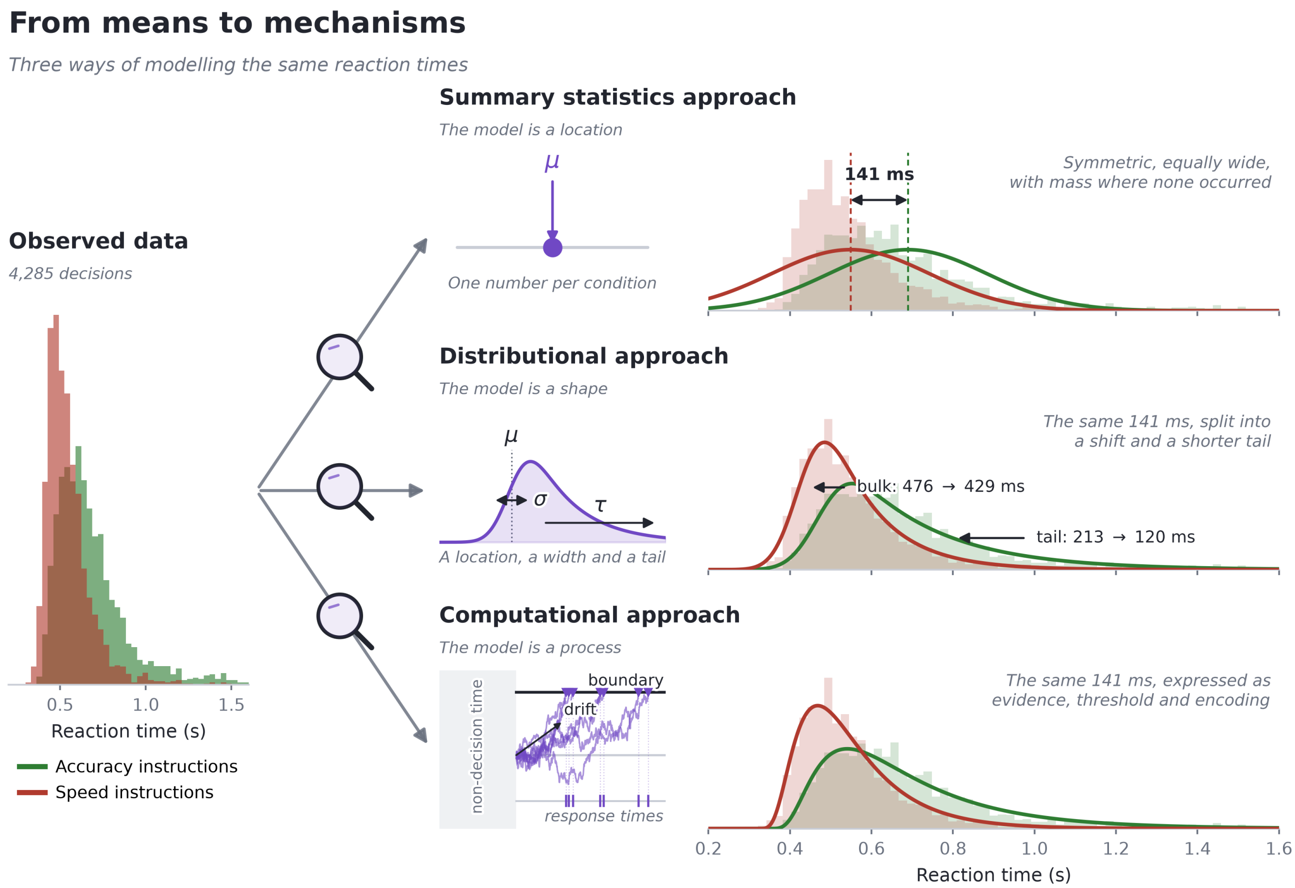



ditionally computed from the same data, making them one of the more promising responses to the "reliability paradox" (Haines et al., 2020; Hedge et al., 2018).

The gap between what a task produces and what is taken from it is widest in neuropsychological assessment, which rests almost entirely on cognitive tasks. The indices extracted from them are total scores, completion times and differences between condition means: quantities inherited for historical rather than theoretical reasons, with no principled mapping onto the constructs they are said to measure (Ratcliff & McKoon, 2022). Older adults, for instance, are reliably slower, but a meta-analysis of diffusion model analyses locates the general age effect on boundary separation and non-decision time, with drift rate differences that are task-dependent and occasionally reversed (Theisen et al., 2021). A latency difference reported as "cognitive slowing" would be more accurately framed as a statement about caution or motor execution. Computational neuropsychology (Steinke & Kopp, 2020) proposes to keep the tasks and replace those scores with parameters that refer to identifiable components of

**Figure 2**

*The three approaches applied to the same trials: correct responses from three participants in the lexical decision data of Wagenmakers et al. (2008), used again in Examples 1 and 2. Observed reaction times are on the left; each row then shows what one model is made of and the distribution that follows from it, drawn back over the same data. The 141 ms difference between instruction conditions is first a single number, then a shift plus a shorter tail, then a statement about how evidence was accumulated. The ex-Gaussian and shifted Wald curves both use the posterior means reported in Example 2.*




processing, so that a deficit is characterized rather than merely detected.

The same holds for brain-behaviour correlations, a field burdened by weak associations and low reliability. Brain-wide association studies based on basic behavioural indices (such as mean RTs) need thousands of participants before their effects replicate (Marek et al., 2022), and the test-retest reliability of common task-fMRI measures is typically low (Elliott et al., 2020). In contrast, parameters derived from cognitive models are more reliable (Haines et al., 2020; Rouder & Mehrvarz, 2024) and closer to what a neural account is about. The threshold adjustment made under time pressure is a case in point: it is a model parameter rather than an observable, and seems to accurately track striatal and pre-supplementary motor activation, and cortico-striatal connection strength, across individuals (Forstmann et al., 2008, 2016; Wiecki et al., 2015). Whether the target is a clinical score or a brain-behaviour correlation, the root cause is the same: we only get what we fit, and fitting means with linear models is a self-imposed limitation.

### The obstacle is practical

Uptake of alternative approaches remains unfortunately modest, and the obstacle is mostly practical. The software landscape for fitting alternative models is fragmented. Diffusion models are fitted in HDDM or HSSM in Python (Fengler et al., 2026; Wiecki et al., 2013), in fast-dm (Voss & Voss, 2007), in DMAT (Vandekerckhove & Tuerlinckx, 2008), or in PyDDM (Shinn et al., 2020); racing accumulator models in DMC and EMC2 in R (Heathcote et al., 2019; Stevenson et al., 2026), or with `SequentialSamplingModels.jl` in Julia (Fisher & Makowski, 2024); bounded and ordinal responses in yet another set, such as `ordbetareg` (Kubinec, 2025). Each toolbox has its own syntax for specifying which parameters depend on which conditions, its own parameterizations, defaults, and its own output objects.

As a consequence, comparing families is difficult and therefore rare. Deciding between an ex-Gaussian and a shifted LogNormal, or asking whether a Weibull distribution would do as well as either, means re-fitting in different packages under different conventions, which slows exploratory work and leaves the choice of likelihood to be inherited from disciplinary convention rather than tested against the data at hand. Second, these toolboxes often sit apart from the regression workflow psychologists already use, such as `brms` in R (Bürkner, 2017), which means missing on the expressive power of that workflow. Every design has to be restated in a toolbox-specific syntax, and features like complex random effects specifications, smooth terms, monotonic effects for ordinal predictors, or post-processing tools, such as marginal effects and means, are effectively out of reach (Bürkner, 2018; Wood, 2017).

`cogmod` aims to address these limitations by facilitting the application of these models inside a framework researchers already use. Every model is defined as a `brms` custom family (Bürkner, 2017), and therefore a Stan model (B. Carpenter et al., 2017) specified with ordinary `brms` formula syntax. Everything a `brms` user already knows then applies unchanged to a Drift Diffusion Model or an ordered-Beta rating model: distributional regression on every parameter, multilevel structure, priors, convergence diagnostics, posterior predictive checks, leave-one-out cross-validation (Vehtari et al., 2017). The `easystats` ecosystem (Lüdecke et al., 2019, 2021; Makowski et al., 2019) has been adapted to accommodate these families, providing first-rate post-processing tools.

We describe below the design of the package and the models it provides, then give three worked examples, one per family of models, to illustrate the workflow and applications.

## Design and Implementation

### Models as `brms` custom families

`brms` translates an R formula into Stan code. Its *custom family* mechanism allows users to supply their own likelihood: a `customfamily` object declaring the distributional parameters and their link functions, plus a `stanvars` object injecting the Stan functions that evaluate the log-density. `cogmod` packages both for each model, and additionally provides the post-processing methods (`log_lik_*`, `posterior_predict_*` and `posterior_epred_*`) that `brms` requires in order to compute information criteria and generate predictions used in post-processing.

In practice, using a `cogmod` model requires two arguments beyond a standard `brm()` call: `family` and `stanvars`. Two further helpers, `cogmod_priors()` and `cogmod_inits()`, are strictly speaking optional but should be treated as part of the call: the families carry parameters measured in seconds and parameters that `brms` cannot recognise by name, and the outlier mixture described below has a degenerate mode that a prior has to fence off. Everything in the package is stated in **seconds**, and nothing rescales itself, so RTs in milliseconds must be divided by 1000 before fitting.

```
library(cogmod)
library(brms)

# Specify the formula using brms' bf()
f <- bf(
  RT ~ Condition +
    (Condition | Participant),
  sigma ~ Condition +
    (Condition | Participant),
  ndt ~ Condition +
    (Condition | Participant),
  poutlier ~ 1,
  family = cogmod_lognormal()
)

# Fit the model
m <- brm(
  f,
  data = df,
  prior = cogmod_priors(f, df),
  init = cogmod_inits(f, df),
  stanvars = cogmod_stanvars(f),
  backend = "cmdstanr"
)
```

Everything else is `brms`. The formula syntax is the usual one, so any parameter can be predicted by covariates, given random effects, splines, autocorrelation structures or monotonic effects. The fitted object is a plain `brmsfit`, so `summary()`, `pp_check()`, `loo()`, `emmeans`, `bayesplot` and the `easystats` functions work out of the box.

Moreover, in most specialized software, the analyst declares which parameters are "free" and which are fixed, and condition effects enter through a toolbox-specific design matrix. Here, each distributional parameter is a regression target with its own formula. Testing the effect of a manipulation on various parameters becomes trivial, as is giving each participant their own parameters through random effects. This allows the location-scale modelling advocated by Williams et al. (2021), in which the dispersion of a participant's RTs is a person-specific quantity with its own predictors, to be the default rather than a special case.

### Relation to existing software

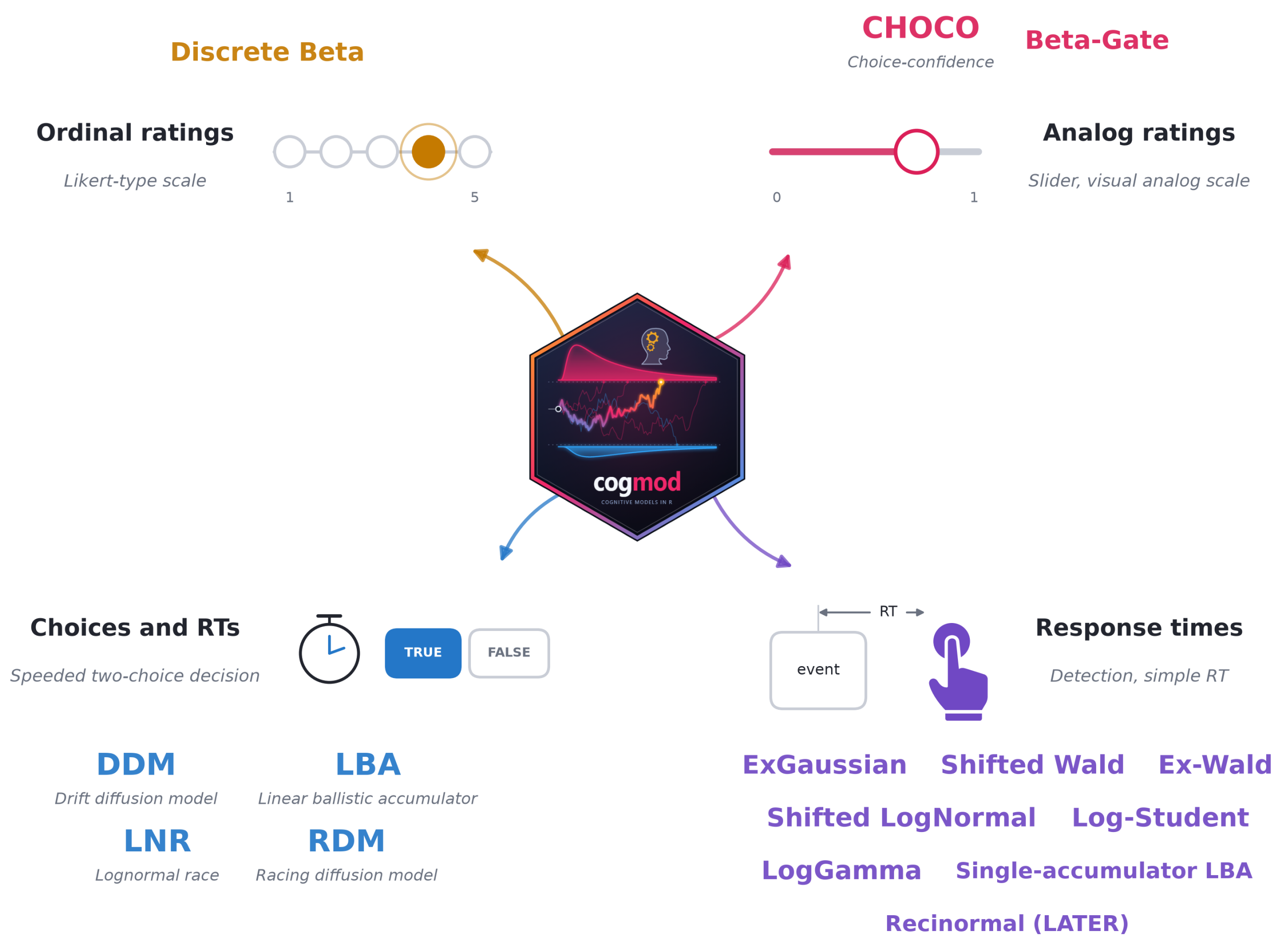
Figure 3 shows the families available at the time of writing. As adding a new model is a self-contained piece of work, the set is expected to grow to track cutting-edge advances in computational psychology. The package's documentation website contains the list of available models, with each family's distributional parameters, link functions and defaults, as well as hands-on tutorials. Because that set is still moving, the family constructors share a single `cogmod_*` naming scheme, and any name that has been retired remains available as an exact synonym so that existing scripts and stored fits keep working.

`cogmod` is not a replacement for the specialized evidence accumulation toolboxes, and the division of labour is worth stating. EMC2 (Stevenson et al., 2026) and HSSM (Fengler et al., 2026) provide sampling algorithms tailored to the geometry of accumulator models, hierarchical structures designed for them and, in HSSM's case, likelihood approximation networks that make otherwise intractable variants estimable. For a study whose scientific question *is* the architecture of the decision process, those are the right tools, and they will be faster and more capable at the far end of model complexity.

`cogmod` targets a different (and arguably more common) situation: a researcher with an experimental dataset, a regression-shaped question and a need for a likelihood that respects the data. For that user the relevant comparison is between a linear model and something better, not between one accumulator toolbox and another. Moving from `brm(RT ~ Condition)` to a shifted LogNormal or a Wald model means changing two arguments rather than changing software, and at every step the analysis remains a regression, with random effects, contrasts and marginal means available as usual.

**Figure 3**

*Response formats covered by `cogmod` and the main families available for each at the time of writing; the package website carries the current list. Six further descriptive RT families are also available (`cogmod_weibull()`, `cogmod_logweibull()`, `cogmod_invweibull()`, `cogmod_gamma()`, `cogmod_invgamma()` and `cogmod_bisa()`) and are compared against the others in Example 2. Every family comes with a `_stanvars()` function supplying the corresponding Stan code, and with `d*()` and `r*()` functions for density evaluation and simulation.*




## Example 1: Five Decision Models, One Workflow

To illustrate the typical workflow of fitting evidence accumulation models, we will fit them to the lexical decision task data of Wagenmakers et al. (2008) (Experiment 1), distributed in the `rtdists` package as `speed_acc`: participants classified letter strings as words or non-words under instructions emphasising either **speed** or **accuracy**. Responses slower than 2 s were excluded. This example uses three participants ($n$ = 4,620 trials, 7.3% of them errors), pooled without random effects for clarity - a simplification we return to at the end of the section.

In order to fit a basic Drift Diffusion Model (DDM) in `cogmod`, one needs to specify 1) a formula for each of the parameters, 2) the model family (cogmod_ddm()) and attach the custom family Stan code with cogmod_stanvars(). As it is common in the field, the "decision" being modelled is **correct vs. error**, which is added to the formula with a dec() term on the response side of the formula. Everything else is the ordinary brms syntax:

```
data(speed_acc, package = "rtdists")
df <- data.frame(
  Condition = unname(c(accuracy = "Accuracy", speed = "Speed")[
    as.character(speed_acc$condition)]),
  RT = speed_acc$rt,
  Error = as.integer(as.character(speed_acc$response) !=
                       as.character(speed_acc$stim_cat)),
  Participant = as.integer(as.character(speed_acc$id))
)
df <- df[df$Participant %in% c(1, 2, 3) & df$RT <= 2, ]

f <- bf(
  RT | dec(Error) ~ Condition,
  boundary ~ Condition,
  bias ~ 1,
  ndt ~ 1,
  poutlier ~ 1,
  sigmadrift = 0, sigmabias = 0, sigmandt = 0,
  family = cogmod_ddm()
)

m_ddm <- brm(f, data = df,
             prior = cogmod_priors(f, df),
             init = cogmod_inits(f, df),
             stanvars = cogmod_stanvars(f)) |>
  add_criterion("loo")
```

The first unnamed parameter, referred to as mu in the Stan code as per brms convention, actually corresponds to the drift rate. The other parameters correspond to boundary separation (boundary), starting point bias (bias), and non-decision time (ndt), respectively. Estimating non-decision time is a common source of difficulty in fitting diffusion models: a shifted distribution assigns zero density to any response faster than ndt, which puts a hard wall in the likelihood at the fastest observed response. Every family that carries an ndt therefore mixes in a small contaminant component, of weight poutlier, that keeps early responses at positive density. The wall becomes a finite cost, and ndt is estimated directly in seconds rather than relative to a sample statistic. The contaminant is a half Normal with a fixed scale of 0.2 s, deliberately lighter-tailed than any decision density in the package: a heavier component eventually explains slow responses better than the model does, and drags ndt up behind it. Two helpers make the mixture inspectable rather than merely assumed - p_outlier() returns each trial's posterior probability of having come from the contaminant, and with_outliers() switches predictions between the decision process alone and the fitted mixture. On these data the mechanism is visible trial by trial: of the 4,285 correct responses modelled in Example 2, the four faster than 200 ms are assigned to the contaminant with probability 1.00, the response at 329 ms with probability 0.51, and by 353 ms the probability is 0.008, against a mean of 0.0015 across all trials. The model discriminates by evidence rather than by a cutoff, which is what having the component buys over trimming the same trials by hand. Not every family is built this way, and the distinction follows the parameter: the descriptive cogmod_exgaussian() of Example 2 has neither ndt nor poutlier, since its exponential component already absorbs the shift, and nor do the rating families of Example 3.

The three sigma* arguments fix the between-trial variabilities of the full seven-parameter diffusion model (Ratcliff & McKoon, 2008) at zero, giving the "simple" four-parameter DDM, which is a common simplification, since the variability parameters are expensive to sample and notoriously hard to recover (Boehm et al., 2018). Setting them to exactly zero is also cheaper than a prior concentrated near zero, because the Stan code falls back to a dedicated density rather than to numerical quadrature. The effect of Condition is also estimated for the boundary separation, which is the textbook hypothesis for a speed-accuracy manipulation, and the results support it: boundary separation drops sharply under speed instructions (95% CI on the linear predictor $[-0.65, -0.55]$). The drift rate also falls, by a smaller and less certain amount ($-0.17$, 95% CI $[-0.29, -0.04]$), which is not what the textbook account predicts; with three participants pooled and no random effects, some of that is likely to be between-participant differences with nowhere else to go.

## Four more architectures

Whether that is the right model, however, is a separate question, and answering it means fitting the alternatives, which only involves a few code changes: a different `family`, its `stanvars`, and whichever distributional parameters that family owns. The **DDM-5** is the model above with between-trial variability in drift rate freed (`sigmadrift ~ 1`). The **LogNormal race** (LNR, Rouder et al., 2015), the **linear ballistic accumulator** (LBA, Brown & Heathcote, 2008) and the **racing diffusion model** (RDM, Tillman et al., 2020) replace the single diffusion with two accumulators racing towards a threshold, and differ in how each accumulator gets there: by drawing its finishing time from a LogNormal, by accumulating linearly with between-trial variability in its rate, or by diffusing with within-trial noise alone. In the last two, the start-point range and the threshold increment enter the likelihood only through their sum, so `sigmabias` needs either a weakly informative prior or to be pinned at zero; `cogmod_priors()` supplies the former, and the safer habit when a threshold is to be interpreted is to read `boundary + sigmabias` rather than either alone.

Because each fit is a `brmsfit` carrying a `loo` criterion, comparing the five is one call:

```
loo_compare(m_ddm, m_ddm5, m_lnr,
            m_lba, m_rdm)
```

## The comparison as a diagnosis

The ranking in Table 1 is three participants' worth of evidence and is not the point. What the comparison is *for* is visible in the simple DDM, 111 ELPD points behind the best model. It carries exactly the mechanism the manipulation is believed to act on, `Condition` on the boundary separation, and it recovered the expected effect; the two models that clearly outpredict it have no threshold parameter free to move at all. Its deficit lies on an axis nobody was testing, and a posterior predictive check locates it (Figure 4 A): the model puts its error distribution to the left of the observed errors. In the upper half of the same panel all five models are close, which is the part worth remembering: a model's problem can be invisible if one only looks at the RTs of correct responses.

Reduced to a single statistic (Figure 4 B), the DDM predicts errors 100 ms *faster* than correct responses (95% CI $[-119, -79]$) against an observed $-15$ ms: the direction is right, the magnitude is not. Errors that are barely faster than correct responses, at an unchanged leading edge, are the signature of between-trial variability in drift rate: errors are then contributed disproportionately by the trials that happened to draw a low drift, which lengthens the error tail without moving its onset (Ratcliff & McKoon, 2008). That is precisely the parameter DDM-5 frees, and freeing it is worth 36 ELPD points and closes about a third of the gap on the statistic ($-64$ ms, $[-95, -30]$). The races need no such parameter, since each already carries a separate drift rate for the error accumulator.

The complication is worth stating rather than smoothing over: the model that matches this particular statistic best is the RDM ($-7$ ms, $[-27, +14]$), and the RDM is last in Table 1. A predictive check on one statistic and a global measure of predictive fit are answering different questions, and they need not agree. What the check buys is not a ranking but a *direction* - it says which parameter is missing, which the LOOIC column alone never does.

The lesson is therefore about diagnosis rather than about which family wins. A poor fit had an

**Table 1**

*Leave-one-out cross-validation for five evidence accumulation models fitted to the choices and reaction times of three participants ($n$ = 4,620). ENP is the effective number of parameters; Difference is the difference in expected log predictive density relative to the best model, with its standard error. Time is the median sampling duration per chain (4 chains, 500 iterations, on a consumer laptop). All Pareto k values are below 0.5, so the estimates are reliable.*

| Model | LOOIC | ENP | ELPD | Difference | SE | Time (s) |
|---|---|---|---|---|---|---|
| LBA | −2657.7 | 8.42 | 1328.9 | 0.0 | 0.0 | 665.1 |
| LNR | −2633.0 | 9.86 | 1316.5 | −12.3 | 10.1 | 69.6 |
| DDM-5 | −2507.3 | 10.74 | 1253.7 | −75.2 | 17.3 | 484.9 |
| DDM | −2436.3 | 10.70 | 1218.2 | −110.7 | 19.1 | 241.7 |
| RDM | −2419.4 | 6.96 | 1209.7 | −119.2 | 18.1 | 806.9 |

identifiable cause, the cause pointed at one parameter, and freeing that parameter improved both the fit and the statistic — with the comparison, the predictive check and the re-fit in one session and one syntax. The alternative arrangement, in which each architecture lives in its own package, does not forbid this sequence so much as make it unlikely to be attempted.

### What the architectures cost

Figure 4 C reports the other half of the decision, and it does not rank the models the same way. The LBA is nominally best and takes 665 seconds per chain; the LNR is 12 ELPD points behind it (SE 10.1) at 70 seconds, nearly ten times cheaper — a difference well within noise for a tenth of the cost, which on these data makes the LNR the sensible default. Freeing drift variability is not free either: DDM-5 costs twice the simple DDM. On three participants these are minutes; multiplied by a hierarchical model over a full sample, they are the difference between an afternoon and a week, and this is the boundary at which the specialized samplers of EMC2 (Stevenson et al., 2026) and HSSM (Fengler et al., 2026) earn their place.

Two caveats bound the example. Errors are rare here (335 trials), so the between-trial variability parameters are weakly identified and the comparison of the two DDMs is correspondingly soft; and three participants pooled without random effects, fitted with short chains, is a demonstration rather than a result. Example 2 returns to what that pooling costs.

## Example 2: Reaction Times Without Choices

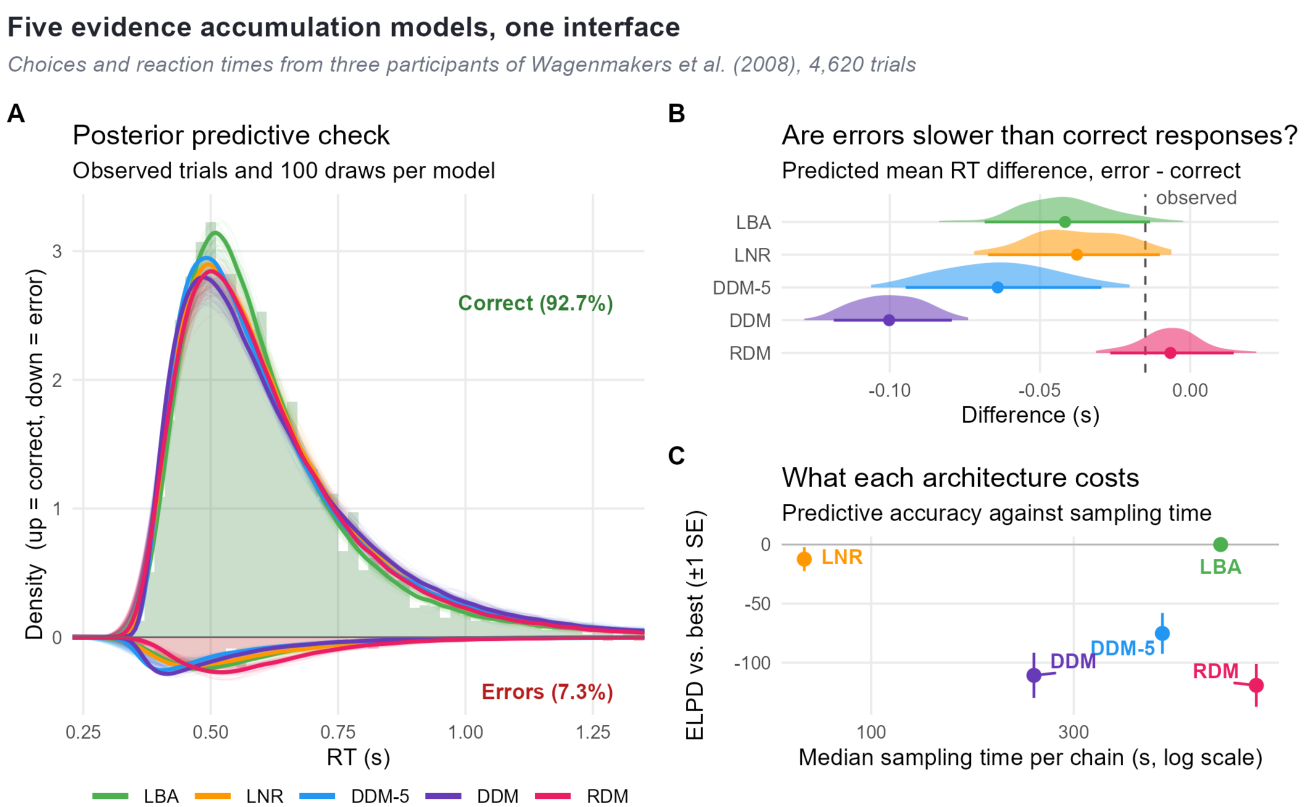
Many datasets carry no informative choice. Accuracy is at ceiling, the task has one response, or errors are too few to estimate anything from — which is the usual situation in the clinical instruments this example ends on. What is left is a duration, and the question becomes which distribution to give it. We keep the same trials as Example 1, now dropping the 335 errors and modelling the correct responses of the same three participants ($n$ = 4,285).

### From a mean to a shape

A Gaussian model gives the familiar answer: responses were 141 ms faster under speed instructions (95% CI $[129, 152]$). Moving to a likelihood built for the shape of reaction times costs one changed argument and two extra formulas:

**Figure 4**

*Five evidence accumulation models fitted to the choices and reaction times of three participants.* ***(A)*** *Posterior predictive check: observed trials as histograms, 100 predictive draws per model as faint lines and their pooled density in bold, with correct responses drawn upwards and errors downwards. Both halves are scaled by the* proportion *of that response, so the area under each curve is its predicted frequency and the check is on the joint distribution of choices and RTs. Draws are taken with* `with_outliers()`*, so the check is on the same mixture the likelihood scores.* ***(B)*** *The same check reduced to one statistic per draw, the predicted difference between the mean error RT and the mean correct RT, against the value observed in the data.* ***(C)*** *Predictive accuracy against sampling cost. Predicted RTs above 3 s (0.05% of draws, produced by near-zero drift rates in the races) are excluded from panels A and B.*




```
f <- bf(RT ~ Condition,
        sigma ~ Condition,
        tau ~ Condition,
        family = cogmod_exgaussian())

m <- brm(f, data = df,
         prior = cogmod_priors(f, df),
         init = cogmod_inits(f, df),
         stanvars = cogmod_stanvars(f))
```

The three parameters reproduce the Gaussian estimate exactly ($476 + 213 = 689$ ms under accuracy instructions, $429 + 120 = 549$ ms under speed), but they attribute it: **67% of the 141 ms speed-up is a shortening of the tail**, only 33% a shift in the bulk. A speed instruction does not move every response earlier; it removes the slow ones. That is the whole argument of the distributional approach in one comparison, and it costs nothing to run.

Note that `mu` here is in seconds directly. It is the *location* of the Gaussian component rather than a scale, so it sits on an `identity` link and is free to go negative - which for fast, heavily-tailed data is sometimes where it belongs, with `tau` carrying the mass. Constraining it positive, as `cogmod` once did, distorts exactly the `mu`/`tau` split that makes the decomposition above worth reporting.

## Sixteen families, one call

The ex-Gaussian is a choice among many, and because every `cogmod` model is a `brmsfit`, deciding between them is the ordinary `loo` workflow (Vehtari et al., 2017) rather than a bespoke one. Each fit is the same few lines with a different family, and the comparison is a single call:

```
m_exgaussian <- brm(
  bf(RT ~ Condition, sigma ~ Condition, tau ~ Condition,
     family = cogmod_exgaussian()),
  data = df, stanvars = cogmod_stanvars(f)
) |> add_criterion("loo")

# the other fifteen differ only in `family` and which distributional
# parameters exist to receive `Condition`

loo_compare(m_gaussian, m_exgaussian, m_lognormal, m_wald, m_lba, ...)
```

The ranking itself is one dataset's worth of evidence and is not the point. What matters is that three questions become answerable in an afternoon that would otherwise each require a separate toolchain: whether the Gaussian default is defensible (it is worse than the next-worst family by 1,202 units of expected log predictive density); whether the families the literature settled on are the right ones (four of the top five are the log-Student, the log-Gamma, the inverse Gamma and the single-accumulator LBA, none with any appreciable presence in the RT literature, while the ex-Gaussian - one of the two most reported families in the field - sits fourth from bottom); and what the better fits cost (the ex-Wald (Schwarz, 2001) takes 942 seconds per chain and is outpredicted by the log-Student at a ninth of that, while the shifted LogNormal reaches ninth place in 36 seconds). The last of these is a column in Table 2 rather than an inference the reader must assemble across papers.

The two nested pairs in the table are worth a second look, because they are the kind of comparison this arrangement makes cheap. Freeing between-trial variability in drift rate moves the Wald from 38.8 points behind to 2.2 (`sigmadrift`, the RT-only counterpart of the DDM-5 of Example 1), and pinning the LBA's start-point range at zero - which gives the recinormal, or LATER model of R. H. S. Carpenter and Williams (1995) - costs 3.1 points and cuts the sampling time nearly in half. Both are one edited argument, and both are like-for-like: the nested model shares its likelihood with the general one.

As in Example 1, *where* a model fails is the more informative question, and the posterior predictive checks that answer it come out of the same call as for any `brmsfit`, since `cogmod` supplies `posterior_predict_*` methods for every family — `modelbased::estimate_prediction()` returns draws ready for plotting (Makowski et al., 2020). Figure 5 shows all sixteen families over the observed distributions: the Gaussian, the Weibull and the Gamma, which the table places far behind, miss the peak and the tail in visibly different ways, while the families at the top of the table are hard to tell apart by eye.

Three caveats apply to reading the table. These models pool three participants without random effects, so families that absorb between-participant heterogeneity more gracefully are advantaged; a hierarchical comparison is easy to specify but expensive to fit. The three families flagged in the caption for high Pareto $k$ are exactly the ones whose predictive draws sit worst over the data, which is what a failing importance-sampling diagnostic usually means - their positions should be read as upper bounds. And LOO measures predictive fit, not theoretical adequacy. The plain shifted Wald sits eleventh of sixteen, below six purely descriptive families, while being one of the five in the table whose parameters refer to a process — and even

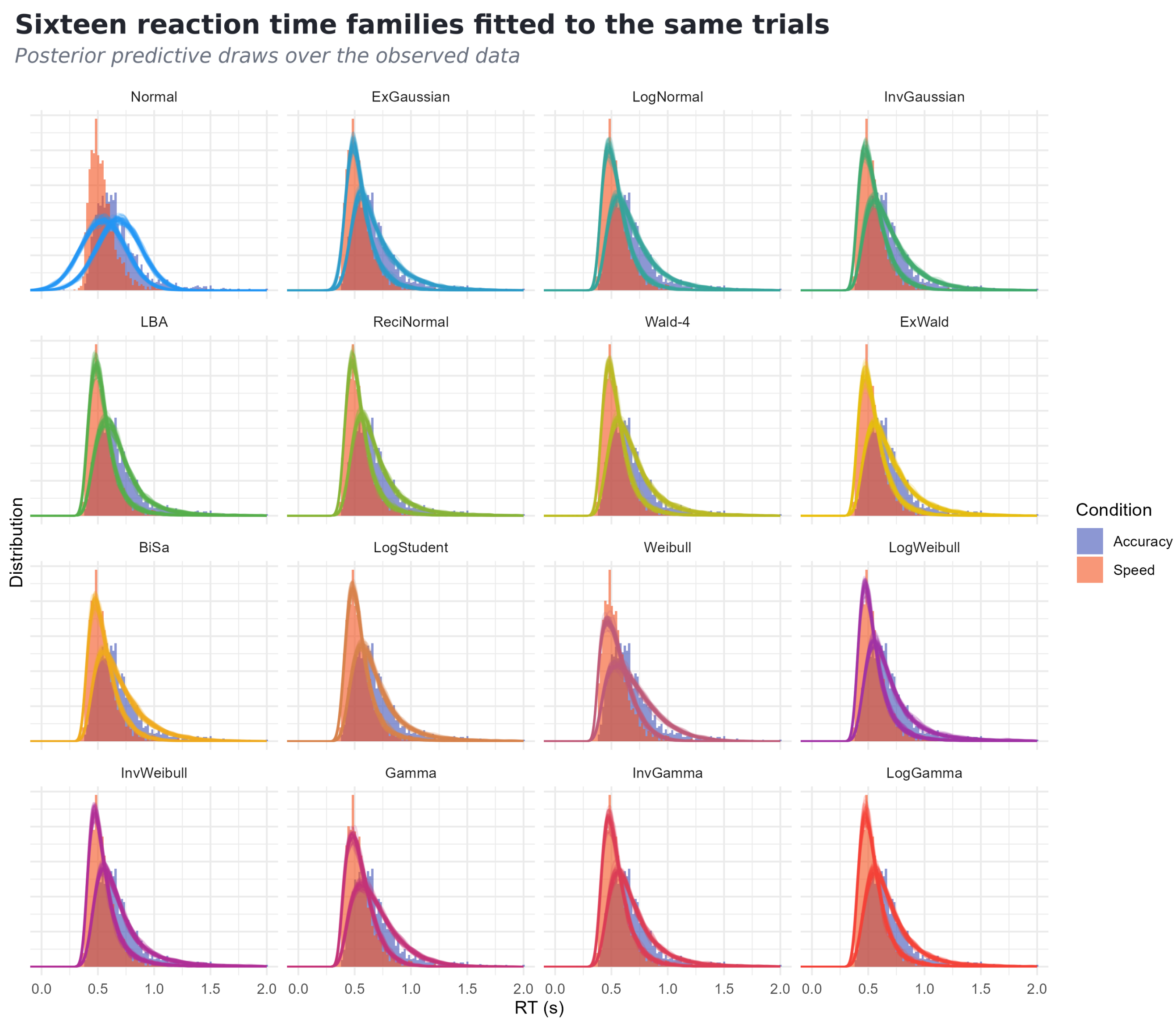
**Table 2**

*Leave-one-out cross-validation for sixteen RT families fitted to the same data (n = 4,285). The LBA here is `cogmod_lba1()`, the single-accumulator variant that predicts a duration only, not the racing model of Table 1; the recinormal is that same family with `sigmabias = 0`. ENP is the effective number of parameters; Difference is the difference in expected log predictive density relative to the best model, with its standard error. Time is the median sampling duration per chain (4 chains, 500 iterations, on a consumer laptop). Three families - the ex-Gaussian, the Gamma and the Weibull - have at least one Pareto k above 0.7, so their positions are approximate and, on the evidence of the diagnostic, optimistic.*

| Model | LOOIC | ENP | ELPD | Difference | SE | Time (s) |
|---|---|---|---|---|---|---|
| Log-Student | −4801.1 | 7.75 | 2400.5 | 0.0 | 0.0 | 101.6 |
| Wald + drift variability | −4796.6 | 7.33 | 2398.3 | −2.2 | 1.6 | 579.1 |
| LogGamma | −4795.0 | 7.37 | 2397.5 | −3.0 | 2.7 | 225.1 |
| Inverse Gamma | −4792.0 | 6.84 | 2396.0 | −4.5 | 3.3 | 143.9 |
| Single-accumulator LBA | −4789.7 | 8.07 | 2394.8 | −5.7 | 2.9 | 370.6 |
| Recinormal (LATER) | −4783.4 | 6.07 | 2391.7 | −8.8 | 2.6 | 199.8 |
| LogWeibull | −4772.7 | 5.41 | 2386.4 | −14.2 | 6.0 | 75.2 |
| Inverse Weibull | −4772.5 | 5.24 | 2386.3 | −14.3 | 6.1 | 93.8 |
| Shifted LogNormal | −4763.3 | 8.42 | 2381.6 | −18.9 | 6.7 | 35.8 |
| Ex-Wald | −4744.8 | 9.69 | 2372.4 | −28.1 | 8.6 | 941.8 |
| Shifted Wald | −4723.5 | 9.35 | 2361.7 | −38.8 | 10.0 | 78.6 |
| Birnbaum–Saunders | −4703.4 | 10.22 | 2351.7 | −48.8 | 11.3 | 56.8 |
| ExGaussian | −4607.9 | 13.45 | 2304.0 | −96.6 | 31.6 | 41.3 |
| Gamma | −4569.6 | 11.85 | 2284.8 | −115.8 | 18.6 | 118.5 |
| Weibull | −4292.3 | 25.96 | 2146.2 | −254.4 | 27.7 | 1267.4 |
| Gaussian | −1888.8 | 7.18 | 944.4 | −1456.1 | 73.8 | <0.1 |

**Figure 5**

*The sixteen families of Table 2, fitted to the same 4,285 trials and drawn back over them. Histograms are the observed RTs in the two instruction conditions; lines are posterior predictive draws. InvGaussian is the shifted Wald, ReciNormal the LATER model, and Wald-4 the Wald with between-trial drift variability freed. Reproduced from the package's* RT-only Models *vignette, which works through the families one by one.*




its parameters have to be read with care here. Placing `Condition` on its boundary separation, the textbook hypothesis for a speed-accuracy manipulation, returns the opposite of the textbook account: a large increase in drift rate under speed instructions (2.67 to 3.83) and no change in boundary separation (95% CI on the linear predictor $[-0.27, 0.13]$). That is a limitation of the model rather than a discovery about the data. A single-boundary Wald sees only correct responses, and with RTs alone drift rate and boundary separation are weakly identified and trade off against each other (Matzke &

Wagenmakers, 2009). Fixing one of the two is the usual remedy; the other is to give the model the errors as well, which is what Example 1 did.

## From a group effect to an individual score

A significant group-level effect and a usable individual-level measure are different things, and a study designed to establish the first can be blind to the second (Hedge et al., 2018). The distinction is decisive for neuropsychology and for differential research, where the point of a task is to produce a number that characterises a *person*, and where a task is often adopted on the strength of a group effect alone. The families just compared are the ones such a task would use, and the question is whether their parameters survive the move from the group to the individual.

We simulated the validation study such a task would need: 30 participants, 80 trials in each of three conditions, RTs generated from a shifted LogNormal process. Two effects were built in deliberately. Condition B produces a small slowing that is nearly identical for everybody (between-participant SD of 0.01 on the log scale). Condition C has a population mean of exactly zero but enormous interindividual variability (SD of 0.30): some participants dramatically slowed, others dramatically speeded.

### *Fitting and reading the model*

The model mirrors the generative process, with random effects on every parameter of the family:

```
f <- bf(
  RT ~ Condition +
    (Condition | Participant),
  sigma ~ Condition +
    (Condition | Participant),
  ndt ~ Condition +
    (Condition | Participant),
  family = cogmod_lognormal()
)
```

The fixed effects reproduce the classic conclusion: a credible effect of condition B ($b = 0.10$, 95% CI $[0.04, 0.17]$, $pd = 99.8\%$) and nothing for condition C ($b = -0.02$, 95% CI $[-0.15, 0.10]$, $pd = 63\%$). The random-effect SDs invert the picture: 0.01 for the condition B slope against 0.28 for the condition C slope, more than twenty times larger. The effect that "worked" is the one on which participants do not differ.

The contrast with the analysis one would normally run is instructive. Computing each participant's mean RT difference by hand yields a between-participant SD of ~31 ms for B − A, though the data were generated with essentially no true variability; a bootstrap shows why, with a per-participant standard error of ~26 ms. By estimating between-participant variability and trial-level noise jointly, the model assigns that spread to the residual term where it belongs and recovers the true SD. This is the model-based counterpart of the noise correction reliability coefficients perform, applied participant by participant rather than to the group as a whole, and it is why model-based individual scores outperform difference scores, especially when participants contribute few, noisy or unbalanced numbers of trials (Haines et al., 2020; Rouder & Mehrvarz, 2024).

### *Quantifying it*

Participant-level estimates are extracted with `modelbased::estimate_grouplevel()`, and their usability as individual scores can be summarised by the **Variance-Over-Uncertainty Ratio** (D-vour), implemented in `performance::performance_dvour()` (Lüdecke et al., 2021):

$$D_{\text{vour}} = \frac{\sigma^2_{\text{between}}}{\sigma^2_{\text{between}} + \mu^2_{\text{within}}}$$

where $\sigma^2_{\text{between}}$ is the observed variability of the participant-level estimates and $\mu^2_{\text{within}}$ their mean squared uncertainty. It is bounded in $[0, 1]$, and values near 1 mean participants differ far more than we are uncertain about where each of them stands. It is close to an intraclass correlation, but its denominator uses the *uncertainty of the estimates* rather than the raw within-participant variance, so that, unlike an ICC, collecting more trials per participant increases it. It is also a normalized version of the signal-to-noise ratio advocated by Rouder and Mehrvarz (2024). The two functions compose directly, so the whole check is `performance_dvour(estimate_grouplevel(m))`.

Table 3 reports the result for every parameter of the model. The ordering is the expected one; the magnitudes are what matter. Condition C reaches 0.93 while condition B collapses to 0.07, far below the 0.58 the same index returns on the raw empirical difference scores, which still contain the trial noise the model has reassigned to the residual. Taken at face value, the empirical value would have marked condition B as borderline usable. The model says it carries no individual-level information at all.

Two features of this table generalise beyond the simulation. First, the most reliable individual-level quantity the task produces is the `mu` intercept (0.94) — *how fast each participant is overall*, which the experimental design treats as a nuisance to be subtracted away. That is the usual situation, not an artefact of this simulation. Second, because `sigma` and `ndt` carry their own predictors and random effects, they get their own reliability verdicts. Here all six are correctly near zero, since every participant was simulated with the same dispersion and the same non-decision time — a useful negative control. Real data often look different: RT variability can discriminate between individuals better than the mean does (Williams et al., 2021), and non-decision time is a natural candidate for a stable person-specific quantity. Had those been the reliable components, the sensible clinical index would have been `sigma` or `ndt`, and not a condition contrast at all.

The recommendation is simple, and a multi-parameter framework is what makes it followable: **compute a reliability index for every parameter intended for use as an individual-level score**, and report it alongside the effect itself.

**Table 3**

*D-vour (Variance-Over-Uncertainty Ratio) for every parameter of the mixed shifted LogNormal model. Values above 0.75 indicate strong group-level differences; values around 0.5 call for caution; values below 0.5 indicate that measurement noise dominates. `estimate_grouplevel()` labels the `mu` block "conditional", since it is the response formula; it is written as `mu` here to match the family's parameter names.*

| Component | Parameter | D-vour |
|---|---|---|
| mu | Intercept | 0.94 |
| mu | ConditionC | 0.93 |
| mu | ConditionB | 0.07 |
| ndt | ConditionC | 0.10 |
| ndt | Intercept | 0.08 |
| ndt | ConditionB | 0.06 |
| sigma | ConditionC | 0.08 |
| sigma | Intercept | 0.06 |
| sigma | ConditionB | 0.03 |

## A clinical reanalysis

The data so far have been a laboratory experiment and a simulation. The case for computational neuropsychology, though, is a case about clinical data, where trials are few, samples are heterogeneous and the index being reported was fixed decades ago. The rest of this example is the same family — a shifted LogNormal with random effects — applied to such a dataset.

We use the UCLA Consortium for Neuropsychiatric Phenomics dataset (Poldrack et al., 2016), openly available on OpenNeuro as `ds000030`, and take the task-switching data from the 42 participants diagnosed with ADHD and the 125 healthy controls who completed it. Participants classified a coloured shape by either its colour or its shape, with the relevant dimension cued on each trial and switching on a quarter of them. Keeping correct responses between 200 ms and 2 s leaves 14,543 trials, at most 96 per participant. That is a realistic clinical quantity, and far below what demonstrations of these models usually assume.

### *What the conventional analysis finds*

The standard index for this task is the **switch cost**, the difference between a participant's mean RT on switch and non-switch trials, read as a measure of executive control. It finds nothing. Controls average 111 ms and the ADHD group 113 ms, $t(82) = -0.13$, $p = .90$. A linear mixed model agrees: a large switch cost overall ($b = 0.11$ s, $t = 13.3$, $p < .001$, so the manipulation worked), no group difference in it ($p = .89$), and no reliable group difference in overall speed ($p = .12$).

No other summary statistic rescues the analysis. Across participants the groups do not differ in mean RT ($p = .16$), RT standard deviation ($p = .21$), coefficient of variation ($p = .96$), or the 10th and 90th percentiles of their RT distributions ($p = .39$ and $p = .18$). The one index that does differ is the skewness of the RT distribution ($p = .02$), which is not a quantity anyone reports. On the conventional reading, this task detects no deficit in this sample.

### *What the model finds*

We fitted a shifted LogNormal in which diagnosis and switching predict the location, the dispersion and the non-decision time, with random effects by participant:

```
f <- bf(
  RT ~ Diagnosis * Switching +
    (Switching | Participant),
  sigma ~ Diagnosis * Switching +
    (1 | Participant),
  ndt ~ Diagnosis + (1 | Participant),
  poutlier ~ 1,
  family = cogmod_lognormal()
)
```

The model agrees about the switch cost: the interaction is null on the location ($pd = 83\%$) and on the dispersion ($pd = 59\%$). The conventional analysis was right about the question it asked. But diagnosis registers on all three parameters at once, and in opposing directions. On the response scale (Figure 6), the ADHD group's decision component is **96 ms slower** (95% CI $[26, 178]$, $pd = 99.7\%$), their non-decision time is **48 ms faster** ($[-75, -16]$, $pd = 99.9\%$), and their log-scale dispersion is lower ($pd = 96\%$). The two effects largely cancel, leaving a total mean difference of 49 ms that still straddles zero ($[-17, 124]$, $pd = 92\%$) — the null the $t$-test reported, arrived at from two credible components pointing opposite ways.

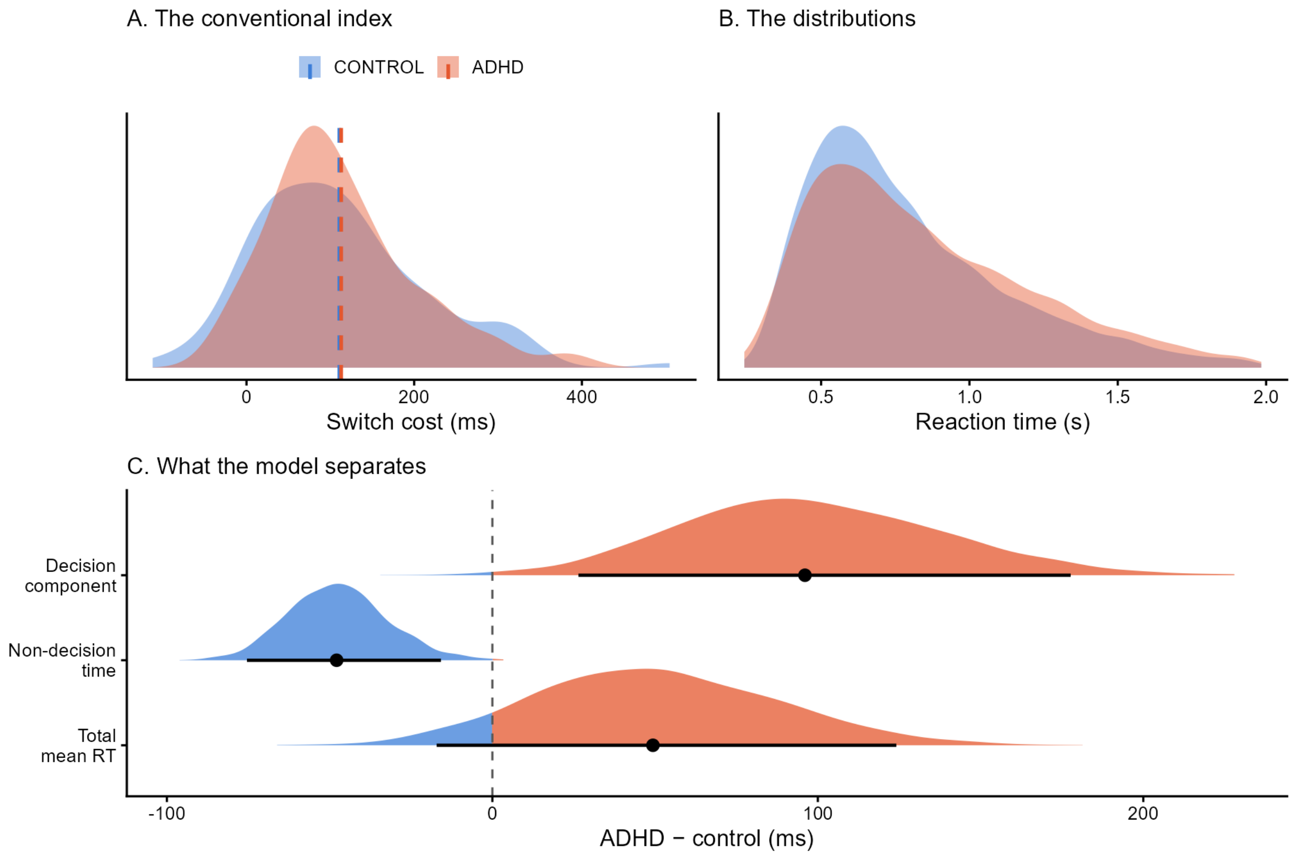
This is the cancellation of Figure 1, arriving in real clinical data: two components move in opposite directions, the mean records their sum, and a difference score computed from means is therefore blind to both. No amount of statistical power on the switch cost would have recovered this, because the switch cost is not where the effect is.

Three caveats bound what this shows. The decision and non-decision parameters trade off in the posterior ($r = -0.44$), the same weak identification that limited the shifted Wald above, so the split between them is less certain than either interval taken alone; what is robust is that a group difference exists in the shape of the distributions and not in their means. The screen of summary statistics reported above is uncorrected, which makes the conventional analysis look, if anything, better than it is. And this is one task in one sample, analysed post hoc: it is a demonstration that the question becomes askable, not a claim about the neuropsychology of ADHD. The substantive point is the one the conventional analysis could not reach — that a task widely read as showing no deficit does distinguish the groups, once the analysis is allowed to say something other than a difference in means.

The two halves of this example are also two halves of one clinical argument. A group difference in a parameter, as here, is a statement about a pop-

**Figure 6**

*The same task-switching data under two analyses (42 ADHD, 125 controls, 14,543 trials).* ***(A)*** *Per-participant switch costs, the conventional executive-control index, which does not separate the groups.* ***(B)*** *The RT distributions the index is computed from.* ***(C)*** *Posterior distributions of the group difference under the shifted LogNormal, on the response scale. The total mean difference straddles zero, but it is the sum of a slower decision component and a faster non-decision component, each credibly non-zero.*




ulation; using that parameter as a patient's score requires the check of the previous section, run on the same fit. Both come out of one model, and neither requires the analyst to leave the regression they started from.

## Example 3: The Same Interface for Subjective Ratings

Cognitive modelling is usually discussed in terms of RTs, but nothing in the interface is specific to them, and the two rating quadrants of Figure 3 are written for response formats psychology collects at least as often. Analog (slider) scales are bounded, and responses pile up at the endpoints, because answering "completely agree" is a different kind of act from answering "agree quite a lot". A mean rating cannot tell the two apart.

`cogmod_betagate()` is `cogmod`'s reparameterization of the ordered Beta model (Kubinec, 2023), in which an observed 0 or 1 arises when an underlying continuous tendency passes a lower or upper "gate". Its four parameters are separately interpretable: `mu` is that underlying tendency, `phi` the consistency of the non-extreme responses, `pex` the overall probability of an extreme response, and `bex` the direction of the extremity. As in every other family, each takes its own formula:

```
f <- bf(score ~ Condition,
        phi ~ Condition,
        pex ~ Condition,
        bex ~ Condition,
        family = cogmod_betagate())

m <- brm(f, data = df, stanvars = cogmod_betagate_stanvars())
```

We simulated 1,000 ratings (`rcogmod_betagate()`) from two conditions in which the manipulation raised both the underlying tendency and the inclination to answer at the top of the scale. The mean rating rises from 0.57 to 0.83, and a $t$-test duly returns one number ($t = 17.5$, $p < .001$). Table 4 shows what the four parameters return instead. The underlying tendency does rise, but so does extreme responding, from 10% to 36% of trials, and those extremes shift towards the upper end — nearly half of the condition B responses sit exactly at the ceiling. Consistency is correctly reported as unchanged, a useful negative control. All four parameters recover their generating values.

Beta-Gate is one of several ways to give a bounded response its point masses — the zero-one inflated Beta and the extended-support Beta (Kosmidis & Zeileis, 2024) are others, and each attributes the endpoints to a different mechanism. Choosing between them is the same problem as choosing an RT family, and it is solved the same way: `loo_compare()` and `modelbased::estimate_prediction()` behave here exactly as they did for the accumulator models of Example 1, and the package's *Subjective Ratings* vignette runs the comparison.

The simulation is not the point; the arrangement is. A response format that ordinarily calls for its own specialised package is here one more `family` argument, carrying the same formula syntax, the same diagnostics and the same post-processing as the diffusion model in Example 1.

**Table 4**

*Beta-Gate parameters for 1,000 simulated ratings; posterior means with 95% credible intervals on the response scale, against the values used to generate the data.*

| Parameter | Condition A | Condition B | Generating values |
|---|---|---|---|
| `mu` — underlying tendency | 0.56 [0.55, 0.58] | 0.71 [0.69, 0.74] | 0.55 → 0.70 |
| `phi` — consistency | 3.29 [2.92, 3.68] | 2.82 [2.36, 3.32] | 3.00 → 3.00 |
| `pex` — extreme responding | 0.10 [0.07, 0.13] | 0.36 [0.31, 0.40] | 0.10 → 0.35 |
| `bex` — direction of extremes | 0.55 [0.41, 0.69] | 0.79 [0.69, 0.87] | 0.50 → 0.80 |

## Discussion

### Summary

`cogmod` implements Bayesian cognitive models as `brms` custom families: descriptive RT distributions, evidence accumulation models, and process models for bounded and Likert-type ratings. The design commitment is minimal. The package supplies likelihoods and the post-processing methods that make them first-class citizens of the `brms` ecosystem, and inherits the rest. The particular families it ships today matter less than the arrangement, which is built so that the set can grow without any of the surrounding workflow changing. The three examples show what this buys. Five accumulator architectures fitted to the same choices and RTs are compared in one call, and the simplest of them is not merely ranked but diagnosed: a predictive check names the missing parameter, and freeing it is one line (Example 1). Dropping the choice, sixteen RT families are compared just as cheaply; because every parameter of the chosen one can carry random effects, the reliability of each as an individual-level score becomes a routine check, and in clinical data the same decomposition recovers a group difference that the task's conventional index, and every other summary statistic we computed, entirely misses (Example 2). The same interface then reaches outcomes the RT literature does not usually address, such as bounded ratings (Example 3).

What the three have in common is that **what you fit is what you get**. A likelihood is not only a claim about the shape of the data; it fixes the vocabulary in which any result can be stated. A Gaussian model can report that two conditions differ by 141 ms and nothing else, because a mean difference is the only quantity it owns. An ex-Gaussian attributes that difference to the bulk or to the tail, a diffusion model to evidence, threshold or encoding, and Beta-Gate separates how highly a participant rates something from how readily they answer at the ends of the scale. Fit and interpretability usually improve together along this sequence, though Table 2 is a reminder that they need not: the shifted Wald is outpredicted by six descriptive families whose parameters mean nothing in particular, and the ex-Gaussian, whose decomposition earns its keep in the same example, is outpredicted by seven. Choosing a likelihood is choosing which questions can be asked at all, and the choice deserves to be made rather than inherited.

## Computational neuropsychology

That argument bites hardest in neuropsychological assessment. A neuropsychological test is a cognitive task, so everything above applies to it directly, with higher stakes attached. The quantity a clinician reports — a completion time, a total score, an interference effect — is exactly the aggregate that the opening example shows to be several things at once, and a patient's deviation from a normative sample is therefore a deviation on an index corresponding to no single mechanism. Model parameters turn "impaired on the Stroop" into a claim about which component is impaired, and two patients with the same score but opposite parameter profiles stop being indistinguishable (Ratcliff & McKoon, 2022; Steinke & Kopp, 2020). Ageing shows what this changes in practice (Theisen et al., 2021), and the clinical reanalysis of Example 2 shows the sharper version of the same point: a conventional index can report no group difference at all while two components underneath it differ substantially, because the index is their sum. The logic covers any group whose deficit could be caution, encoding, motor execution or evidence quality.

The reliability check that precedes it supplies the rest of the argument. A parameter is usable as a clinical score only if it separates people by more than it is uncertain about each of them, and this framework makes that checkable for every parameter rather than assumed for whichever one the test happens to report. It also makes visible that the best individual-level quantity a task yields is often not a condition contrast, but a person's overall speed, dispersion or non-decision time — quantities the experimental tradition subtracts away as nuisance and a clinical tradition has no reason to.

The same reparameterization helps on the neural side. Correlating brain measures with mean RT or accuracy asks the brain to explain a quantity that mixes mechanisms; correlating them with drift rate, threshold or non-decision time asks it to explain one at a time, which is why parameters have proven the more tractable targets in model-based cognitive neuroscience (Forstmann et al., 2016; Wiecki et al., 2015). `cogmod` does not replace the joint approaches that estimate behavioural and neural parameters together, but it lowers the cost of the two-stage version. Clinical datasets arrive with covariates (age, education, group, medication) and unbalanced trial counts; here both are handled by the formula syntax used for everything else, and participant-level estimates come out of `estimate_grouplevel()` ready to be correlated with imaging or clinical measures. The obstacle to a computational neuropsychology was never the argument, which has been on record for two decades. It was that the modelling stayed expensive enough to remain a research activity rather than an analysis step.

## Limitations

**Computational cost.** General-purpose HMC on custom Stan code is slower than a sampler built for a specific model class. Table 1 and Table 2 make this concrete: the LBA took between 6 and 10 minutes per chain on datasets of two to four thousand trials without random effects, and hierarchical versions cost substantially more. Users whose primary question concerns the fine structure of the decision process, or who must fit many participants with complex accumulator architectures, are better served by EMC2 (Stevenson et al., 2026) or HSSM (Fengler et al., 2026).

**Identifiability.** The flexibility of the interface makes it easy to write down models the data cannot support. Regressing every parameter of an LBA on every condition is one line of code and usually a bad idea: evidence accumulation parameters trade off against each other, and between-trial variability parameters are notoriously poorly identified (Boehm et al., 2018; Evans & Wagenmakers, 2020). The package does not, and cannot, prevent this. Prior predictive checks, parameter recovery on simulated data and the standard Bayesian workflow (Gelman et al., 2020) remain the user's responsibility; the `d*()` and `r*()` functions provided for every family - and the `p*()` cumulative distribution functions available for the diffusion, racing diffusion and Wald families - exist to make those checks cheap.

**Maturity.** The package is under active development, and the set of families reported here is a snapshot rather than a specification. Parameterizations may still change, and have: `ndt` was until recently a fraction of a user-supplied minimum RT rather than a quantity in seconds, and the ex-Gaussian's `mu` was constrained positive. The package website and `NEWS.md` track the current state, and record what to change in existing code. Several descriptive RT families have little precedent in this literature, and while Example 2 suggests they deserve attention, their behaviour is correspondingly less well documented than the ex-Gaussian's or the shifted Log-Normal's.

**Scope.** `cogmod` currently covers single-stage decisions and rating responses. Reinforcement learning models, sequential-sampling models with more than two accumulators, collapsing boundaries and joint modelling of behaviour with neural measures are not yet available; some are planned, and some are poorly suited to the custom-family mechanism and may never be.

**Analytically tractable likelihoods only.** That last point is the sharpest boundary on what `cogmod` can become, and it follows directly from the design. A `brms` custom family is a Stan model, so every family shipped here needs a likelihood that can be written down in closed form and evaluated cheaply enough for HMC. Several of the models a user might reasonably want next do not have one: the leaky competing accumulator (Usher & McClelland, 2001), diffusion processes with collapsing boundaries or time-varying drift (Hawkins et al., 2015), and richer racing architectures are defined by a generative simulation rather than by a density, and where a density does exist it often requires numerical integration, which gradient-based sampling tolerates poorly. The general remedy is to stop requiring the likelihood. Simulation-based inference (Cranmer et al., 2020) trains a neural network on data simulated from the model, either to approximate the likelihood — the route taken by the likelihood approximation networks that make otherwise intractable variants estimable in HSSM (Fengler et al., 2021, 2026) — or to amortize the posterior directly, as in BayesFlow (Radev et al., 2020, 2023). That literature is moving quickly and `cogmod` follows it closely; whether such approximations can be exposed behind the same formula interface, so that a simulation-defined model is specified, sampled and post-processed like any other family, is an open question and the direction in which we hope to extend the package.

## Conclusion

The gap between what cognitive data contain and what is extracted from them is large, and it is maintained by friction rather than by disagreement. Few researchers would defend a mean difference as an adequate summary of a reaction time distribution, yet many report one, because the alternative has meant learning a new toolchain for every model worth trying. By implementing cognitive models as families within a regression framework psychologists already use, `cogmod` aims to make the better analysis the easier one; by putting descriptive and generative models under one interface, it aims to make the choice between them an empirical question rather than an inherited convention.

## Open Practices Statement

`cogmod` is open source (MIT licence) and available at https://github.com/DominiqueMakowski/cogmod, with documentation at https://dominiquemakowski.github.io/cogmod/. The lexical decision data of Wagenmakers et al. (2008) are not redistributed with the package; they are read from the `rtdists` package, where they appear as `speed_acc` courtesy of the original authors, and the subset used here is reconstructed by `paper/wagenmakers.R` in the repository. The simulated data used in the Introduction are bundled as `badlm` (with their generating script in `data-raw/`). The clinical data of Example 2 are not redistributed; they are downloaded from OpenNeuro (`ds000030`) by `paper/download_cnp.R` in the same repository, and `paper/example_cnp.R` performs the trial-level extraction described above. All models reported here, the scripts used to fit them, and the code reproducing every figure and table are available in that repository; the two model-comparison tables are regenerated by `paper/make_tables.R` from the fits the package vignettes own, so the paper and the documentation cannot report different numbers for the same model. None of the analyses were preregistered; the Introduction, Example 3 and the reliability section of Example 2 use simulated data, while Example 1 and the rest of Example 2 reanalyse previously published data.

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